AI-Augmented Game Development Workflow for Unity 6

Phase 0: Environment Deployment and AI Interoperability

The foundational phase establishes the technical infrastructure required for a seamless integration between the Unity 6 engine and the AI-driven development environment.

- **Unity 6 Engine Configuration**: Deploying the Unity 6 (6000.x) environment using the Universal Render Pipeline (URP) 2D template ensures compatibility with modern shader graphs and lighting systems. This includes initializing the project with a 2D-specific workflow to support orthographic rendering and specialized 2D light components.
- **Directory Standardization and Asset Pipeline**: Establishing a standardized directory hierarchy (e.g., `Assets/Scripts/`, `Assets/Prefabs/`, `Assets/Arts/`) is critical for maintaining an organized codebase. This structure facilitates clear separation of concerns, allowing the AI to differentiate between logical scripts and visual assets during the development cycle.
- **AI Bridging via Model Context Protocol (MCP)**: Integrating Claude Desktop with the MCP for Unity allows the LLM to achieve local file system interoperability. This configuration enables the AI to read, analyze, and modify C# scripts directly within the project's `/Scripts` directory, significantly reducing manual context switching.

  MCP for Unity: https://github.com/CoplayDev/unity-mcp

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English 简体中文

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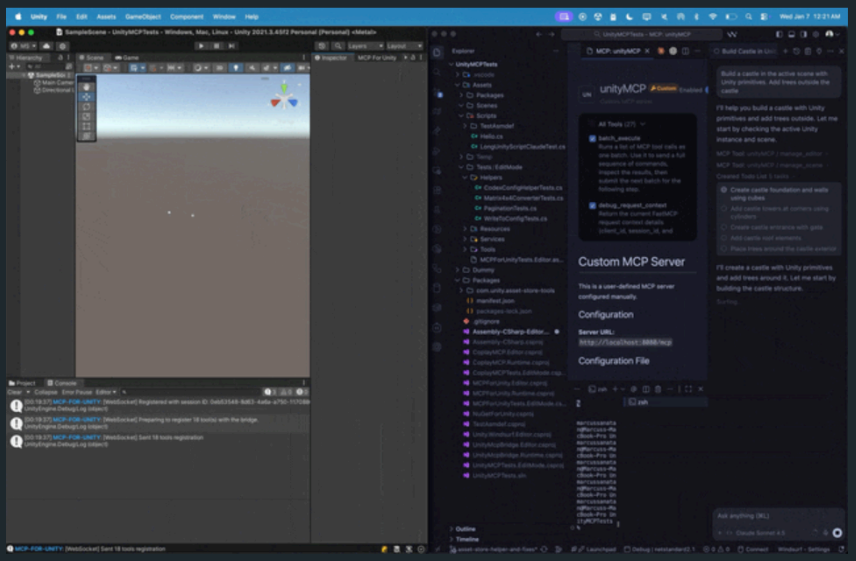
Create your Unity apps with LLMs! MCP for Unity bridges AI assistants (Claude, Claude Code, Cursor, VS Code, etc.) with your Unity Editor via the Model Context Protocol. Give your LLM the tools to manage assets, control scenes, edit scripts, and automate tasks.



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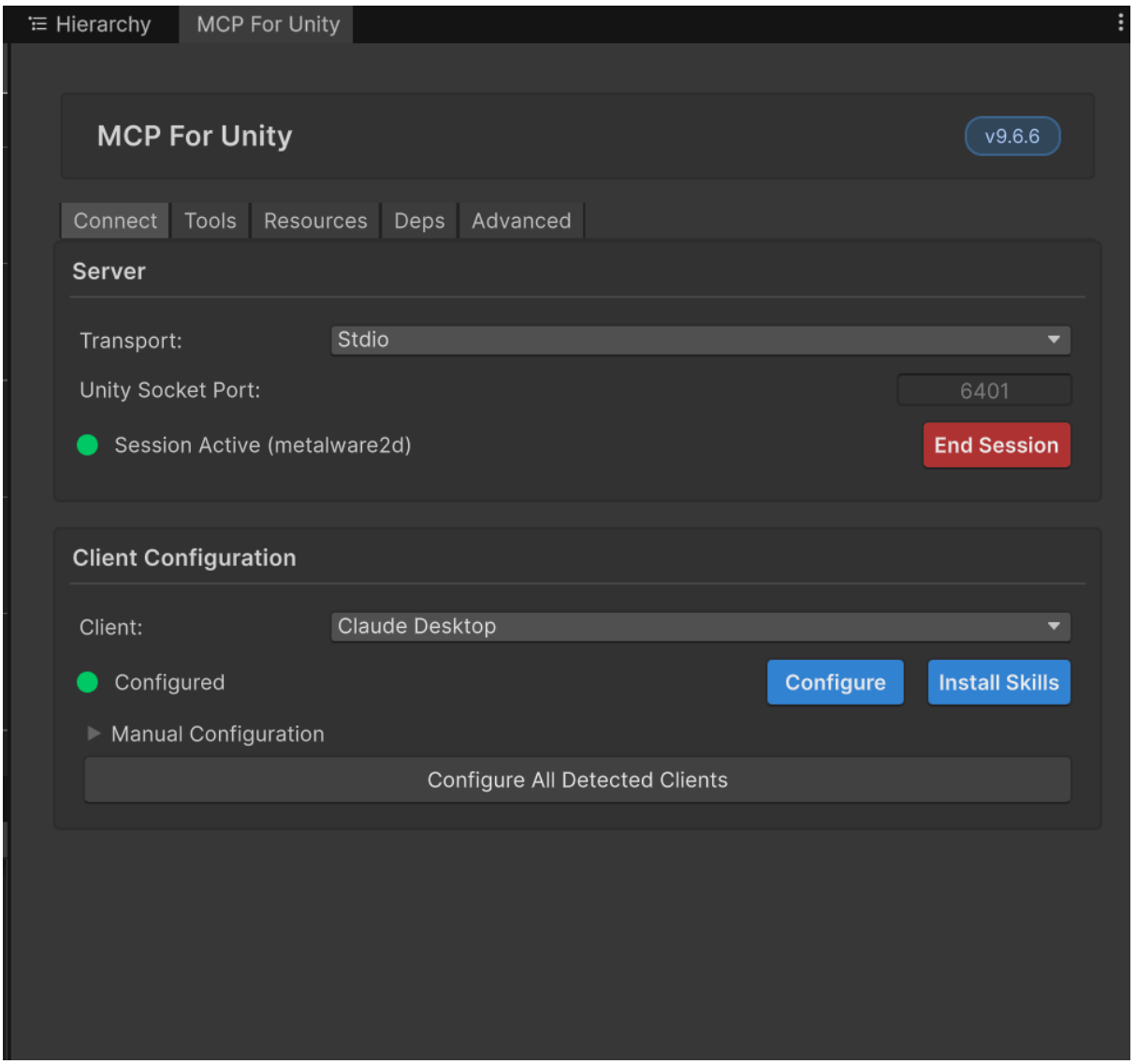
- 
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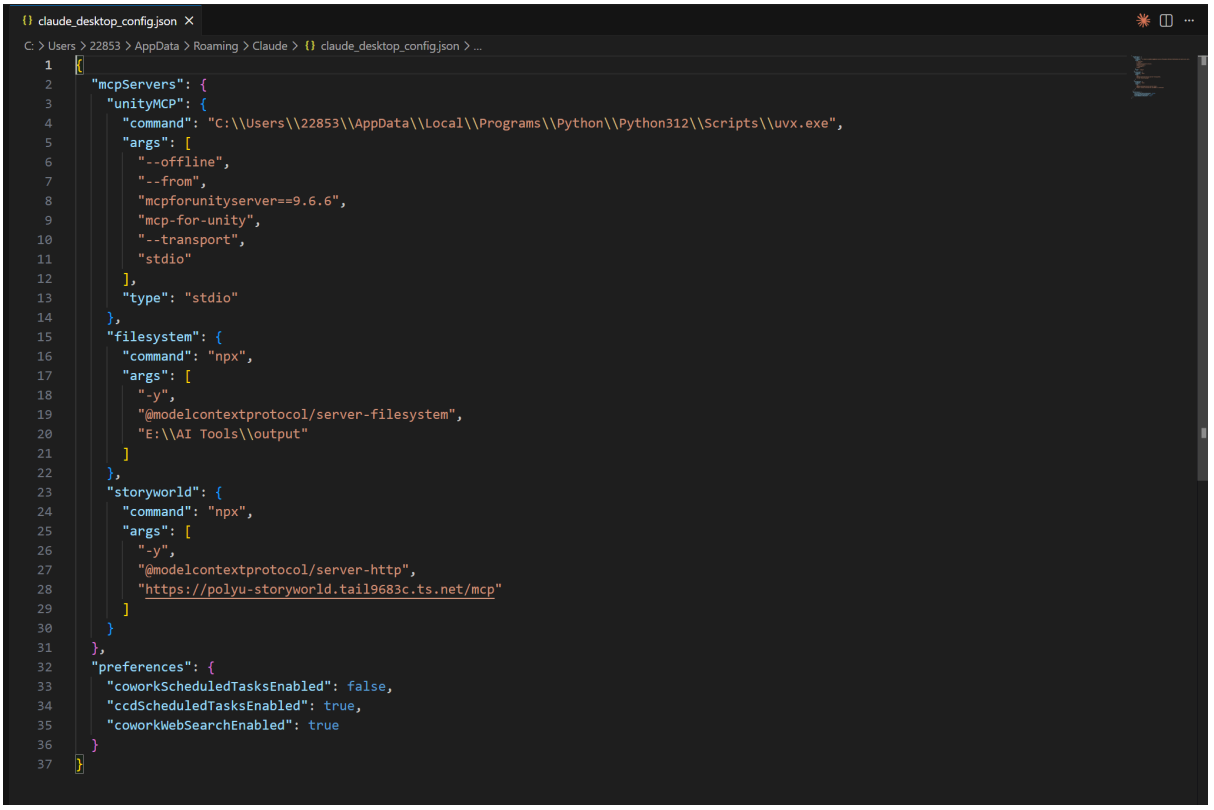


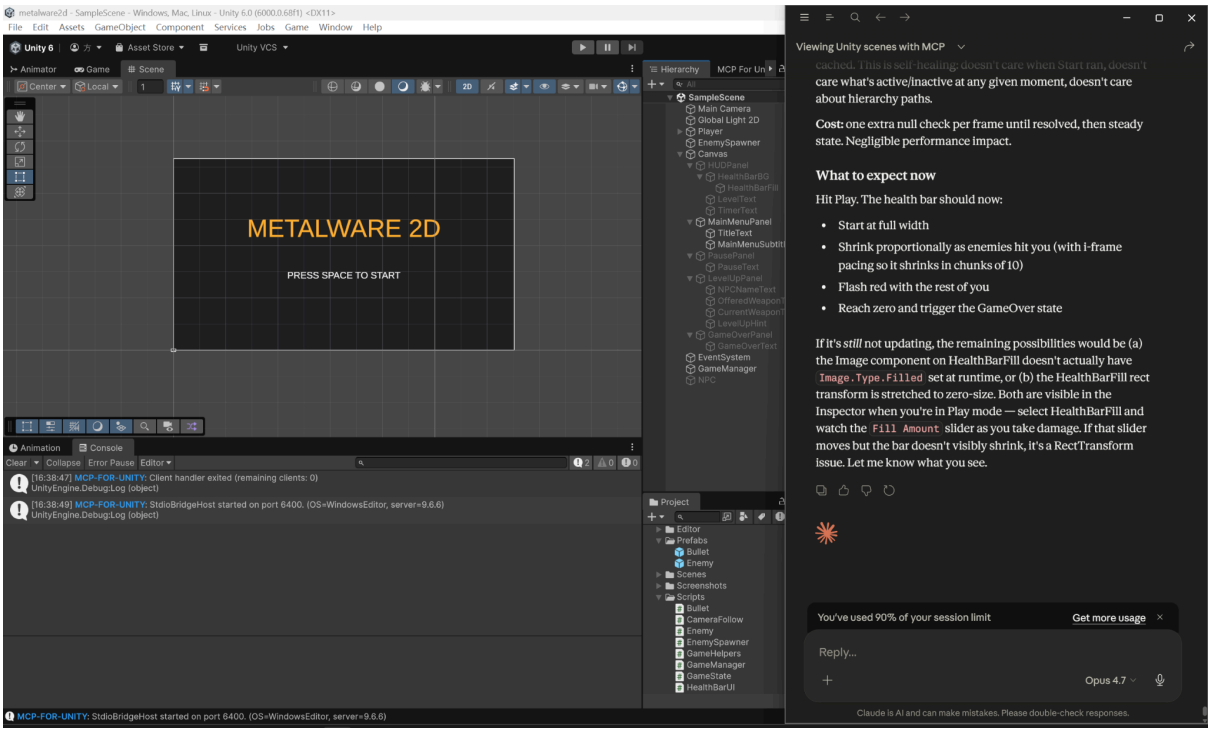
- **Contextual Knowledge Synchronization**: Uploading high-level architectural summaries—such as the project's core gameplay functional features and asset maps—into the AI's long-term memory ensures that generated code adheres to existing singleton patterns and event-driven architectures.
- **Dependency Management**: Ensuring that essential Unity packages, such as the New Input System and TextMeshPro, are pre-installed in the Project Manager. This allows the AI to utilize modern API calls rather than deprecated legacy systems during the generation of controller and UI logic.

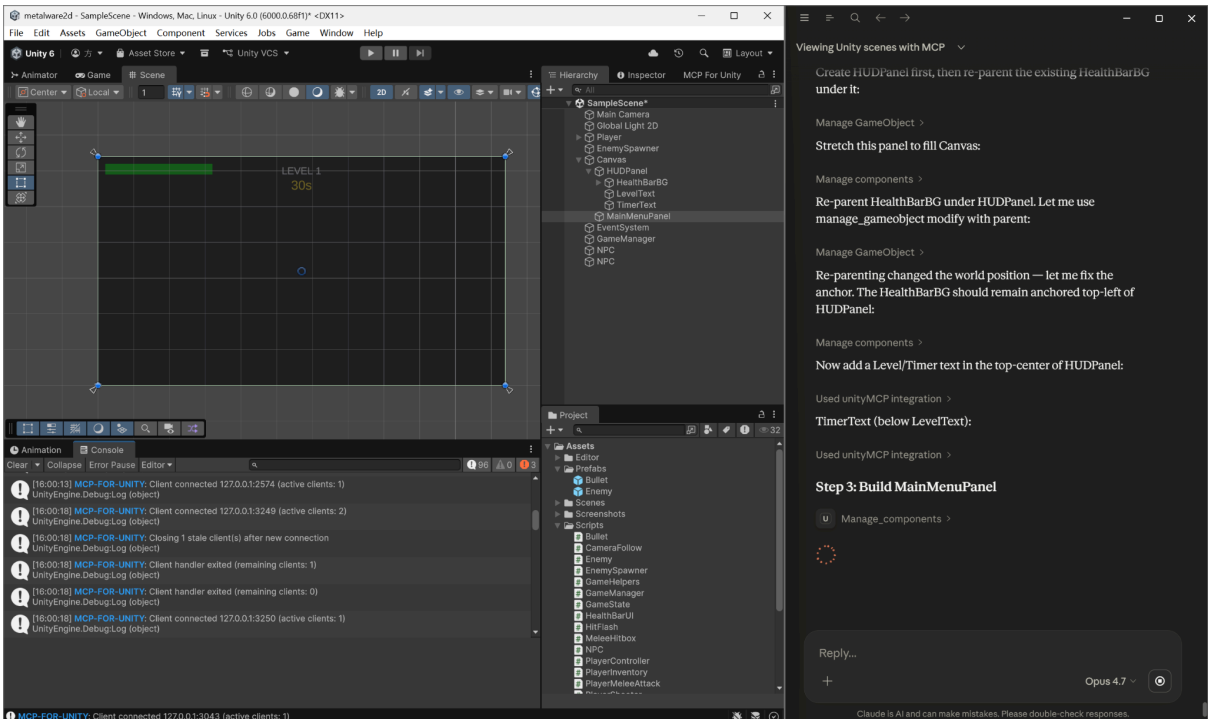
Phase 1: Systems Architecture and Core Infrastructure

During the initialization phase, the focus is on establishing a robust global control framework and directory structure to provide the AI with a coherent context for code generation.

- **Project Initialization**: Deploying Unity 6 with the Universal Render Pipeline (URP 2D) ensures modern rendering capabilities. The orthographic camera is configured with a viewport size of 10 units to optimize the top-down perspective.
- **Centralized State Management**: Implementing a `GameManager` using the Singleton pattern facilitates the governance of game states, including MainMenu, Playing, Paused, LevelUp, and GameOver.
- **Modular Scripting Standards**: Decomposing logic into 19 discrete C# scripts adheres to the Single Responsibility Principle (SRP), which enhances the accuracy of AI-driven code refactoring via MCP.

- **Rendering and Sorting Hierarchy**: Utilizing the URP 2D Light system and defining strict sorting orders—such as assigning the background map to -100—ensures visual consistency and depth management.

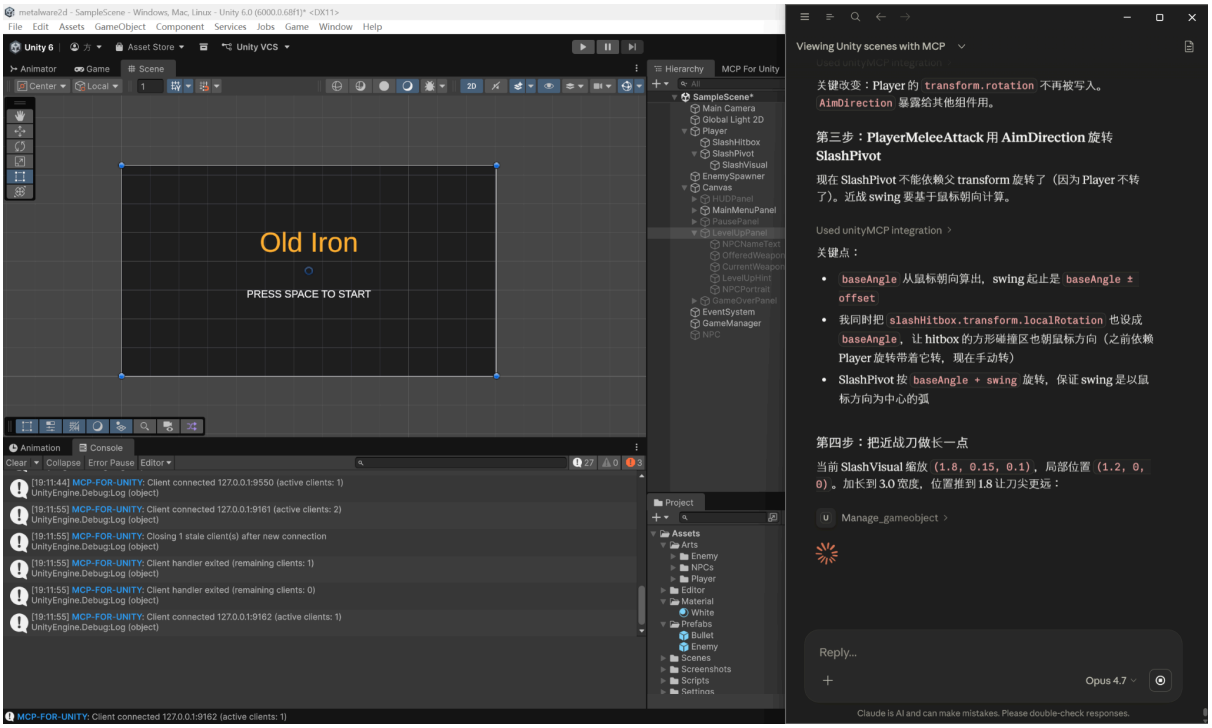




Phase 2: Entity Physics and Combat Mechanics

This phase involves leveraging physics engines and component-based design to construct highly responsive entity behavior models.

- **Kinematic and Dynamic Control**: The player character utilizes `Rigidbody2D` and `BoxCollider2D` for 8-directional movement, incorporating smooth acceleration and deceleration algorithms.
- **Modular Combat Systems**:
    - **Ranged Interaction**: The `PlayerShooter` script manages projectile instantiation, bullet prefab handling, and fire-rate cooldowns.
    - **Melee Interaction**: The `MeleeHitbox` component handles spatial overlap detection to execute damage calculations based on weapon-specific variables.
- **Enemy Intelligence and Spawning**: The `Enemy.cs` script implements pathfinding logic to track player coordinates, while the `EnemySpawner` manages difficulty scaling and randomized wave generation.



Phase 3: Data-Driven Progression and Interaction

The workflow transitions to creating non-linear growth paths by decoupling game logic from numerical datasets.

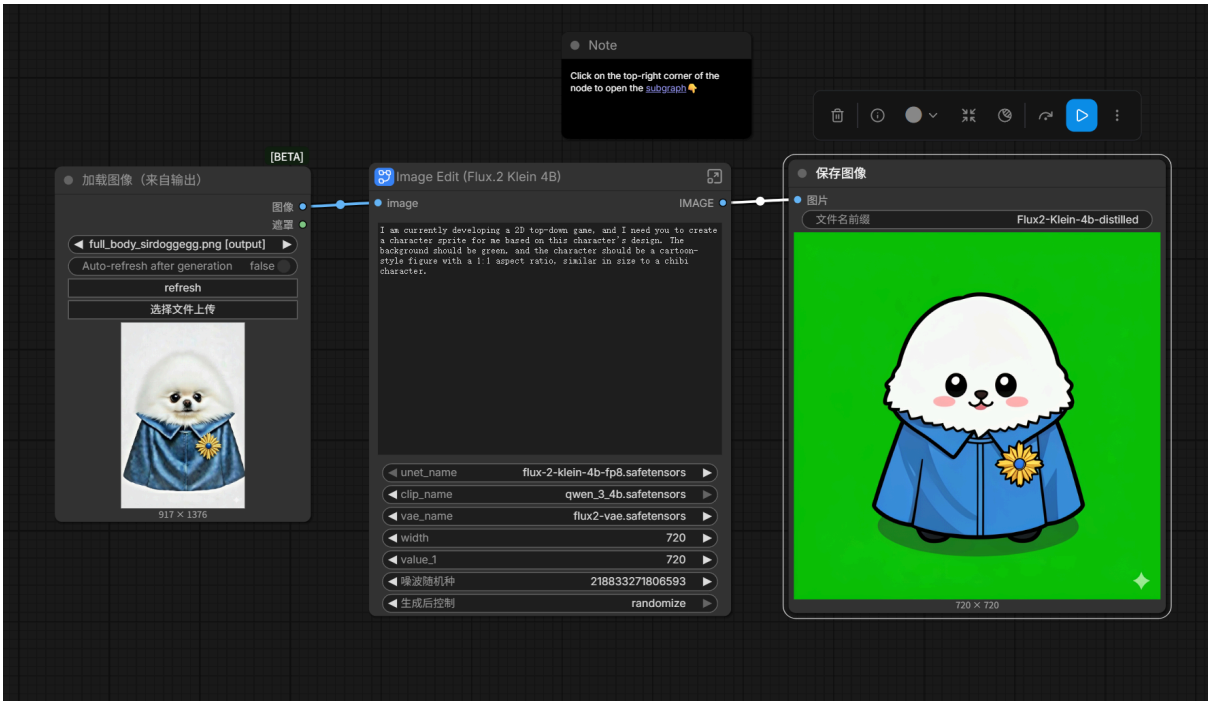
- **Weapon Database Architecture**: A centralized `WeaponDatabase` stores attributes such as damage, fire rate, and range, allowing for rapid balance tuning without modifying core scripts.
- **Interactive Upgrade Framework**:
    - **Trigger Mechanism**: Collisions with NPC entities trigger the `LevelUp` state, which suspends active gameplay to initiate the upgrade interface.

    - **Visual Hierarchy Optimization**: The upgrade UI presents a 420×420px NPC portrait as a focal point, with vertical comparisons between "Offered" and "Current" weapon stats.
    - **Input Mapping**: Strategic player choices are guided by mapped inputs ([E] to Equip, [K] to Keep), which update the `PlayerInventory` in real-time.

Phase 4: Asset Integration and Visual Polishing

Visual assets are integrated through technical configurations that optimize performance and player feedback.

- **Seamless Environment Rendering**: AI-generated 1024×1024 textures are configured with `Wrap Mode` set to `Repeat` and a `Tiled` draw mode to create an infinite background spanning 420×420 world units.
- **Juice and Sensory Feedback**: The `HitFlash` system provides immediate visual confirmation of damage through sprite color manipulation, enhancing the "feel" of combat.
- **Interface Refinement**: TextMeshPro is utilized across all UI panels to maintain high-definition typography, while the health bar system utilizes color gradients and smooth scaling for real-time health visualization.
- **Assets Create:** I got some character images from hugging face repo and using comfyUI to edit these images to game assets.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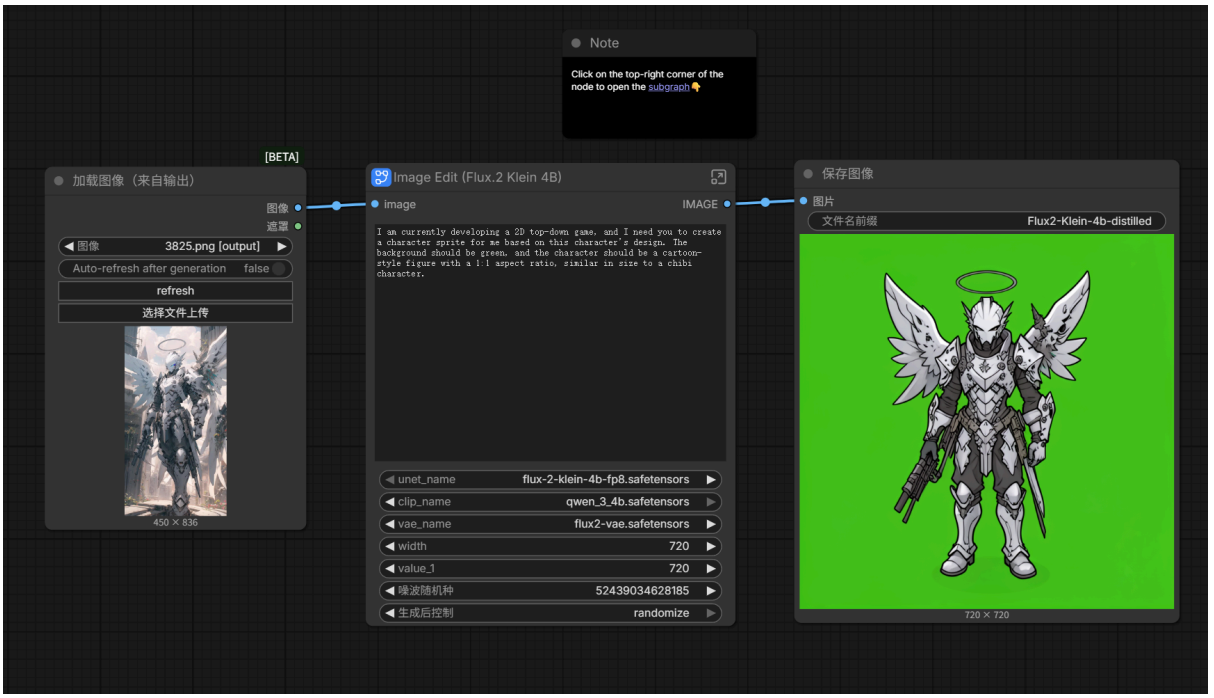
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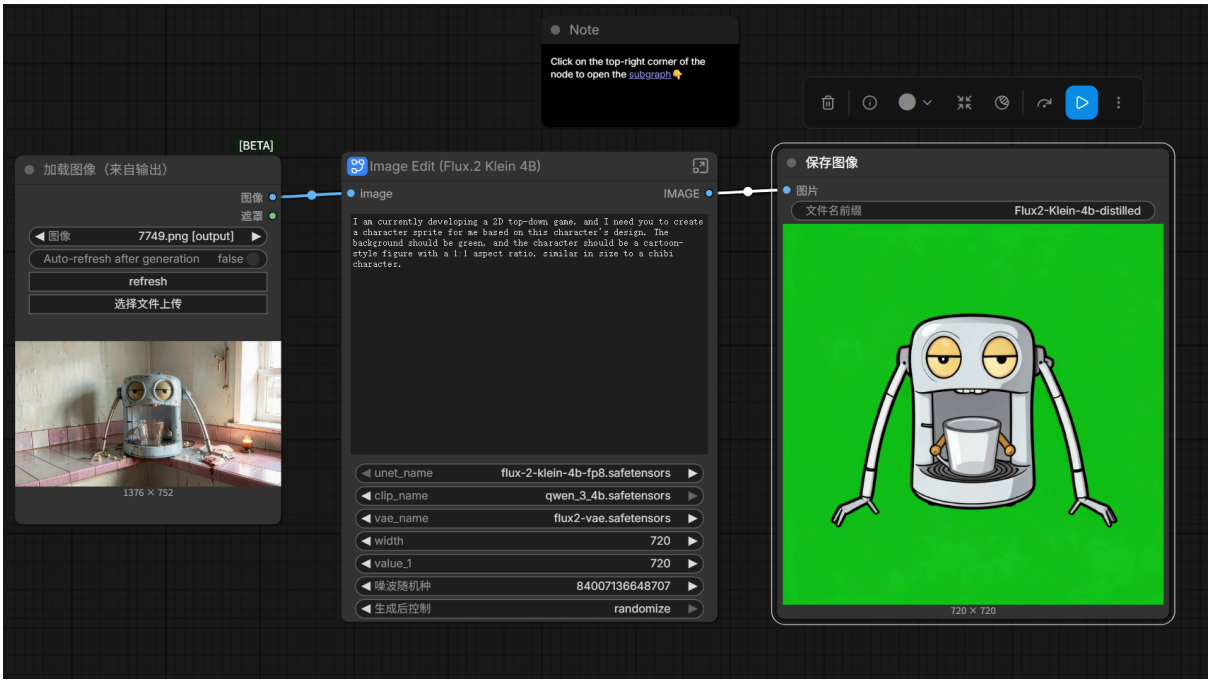
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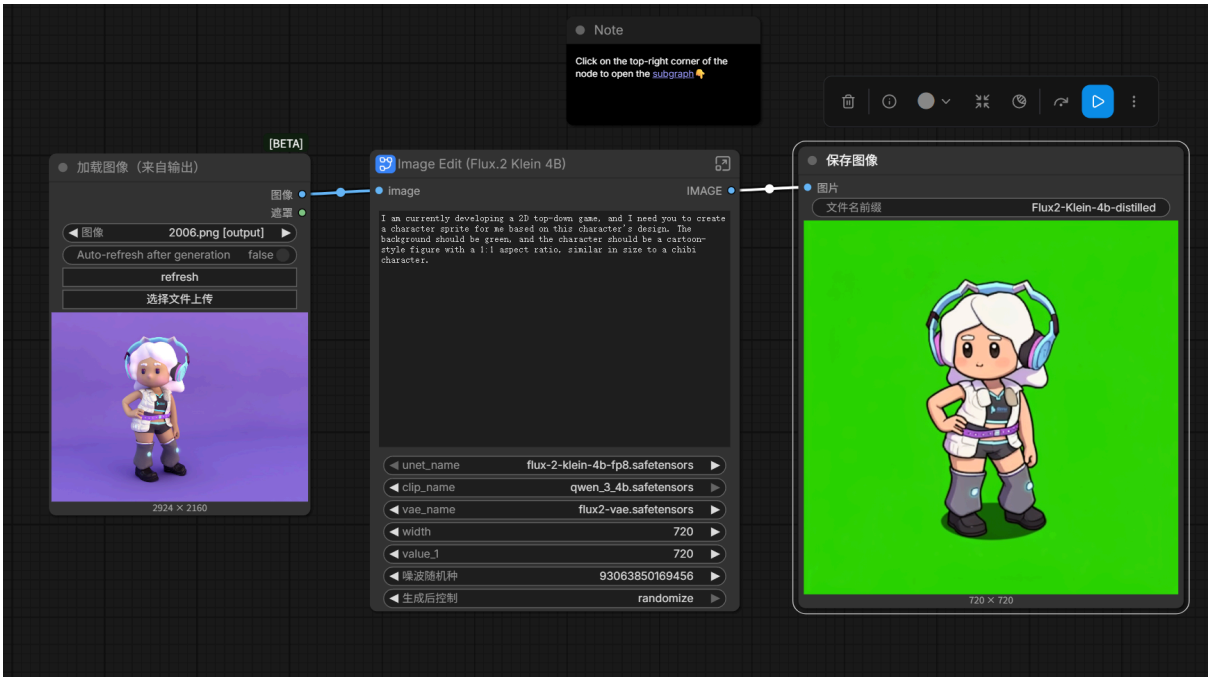

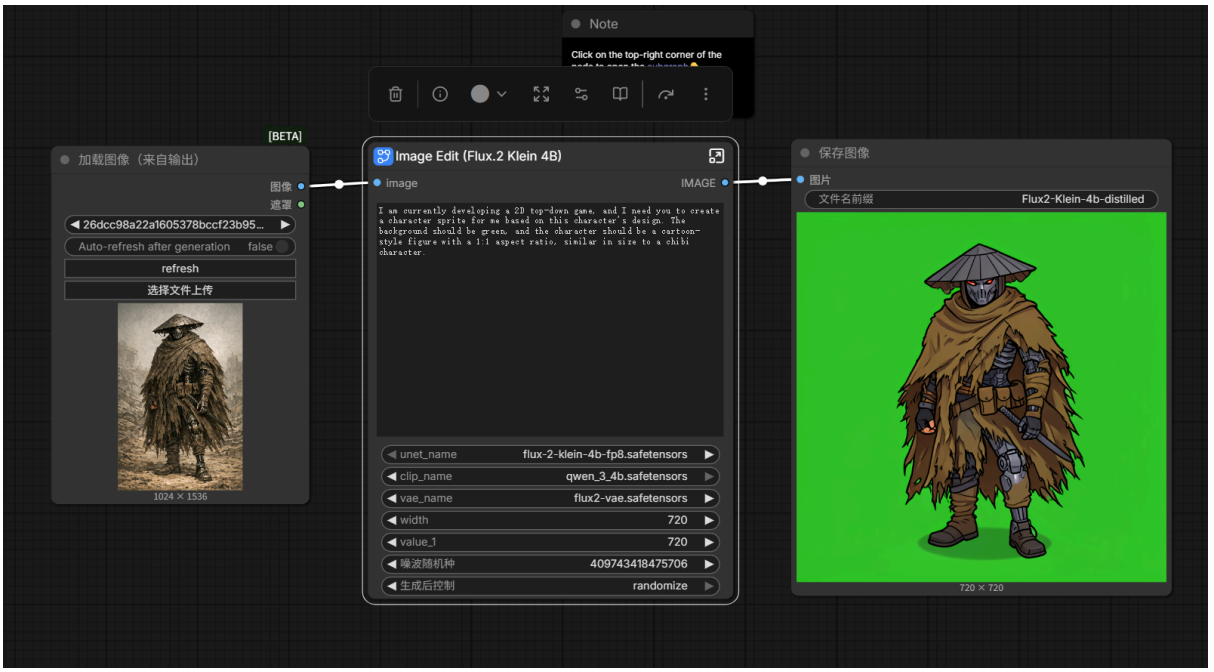

Phase 5: Optimization and Systematic Validation

The final stage ensures the project maintains scalability and performance stability through rigorous engineering practices.

- **Performance Optimization**: The architecture is designed to support Object Pooling for high-frequency entities like bullets and enemies to minimize memory fragmentation and garbage collection overhead.

- **Physics Layer Management**: Explicit Collision Layers and Masks are defined to prevent redundant physics calculations and ensure accurate interactions between projectiles, entities, and environment boundaries.
- **Iterative Debugging**: Tools like `Test.cs` are employed for rapid prototyping of wave difficulty and state transition verification, ensuring the robustness of the core gameplay loop.